

BOND BASICS

Bond and shares form part of the *capital markets*. Shares are *equity capital* while bonds are *debt capital*. So bonds are a form of debt, much like how a bank loan is a form of debt. Unlike bank loans however bonds can be *traded* in a market. A bond is a debt capital market instrument issued by a borrower, who is then required to repay to the lender / investor the amount borrowed plus interest, over a specified period of time. Bonds are also known as *fixed income* instruments, or *fixed interest* instruments in the sterling markets. Usually bonds are considered to be those debt securities with terms to maturity of over one year. Debt issued with a maturity of less than one year is considered to be *money market* debt. There are many different types of bonds that can be issued. The most common bond is the *conventional* (or *plain vanilla* or *bullet*) *bond*. This is a bond paying regular (annual or semi-annual) interest at a fixed rate over a fixed period to maturity or redemption, with the return of *principal* (the par or nominal value of the bond) on the maturity date. All other bonds will be variations on this.

The different types of bonds in the market reflect the different types of issuers and their respective requirements. Some bonds are safer investments than others. The advantage of bonds to an investor is that they represent a fixed source of current income, with an assurance of repayment of the loan on maturity. Bonds issued by developed country governments are deemed to guaranteed investments in that the final repayment is virtually certain. In the event of default of the issuing entity, bondholders rank above shareholders for compensation payments. There is lower risk associated with bonds compared to shares as an investment, and therefore almost invariably a lower return.

We can now look in more detail at some important features of bonds.

B. Description

We have said that a bond is a debt instrument, usually paying a fixed rate of interest over a fixed period of time. It is a collection of cash flows, as illustrated at figure 2.1. This shows an hypothetical example, where the bond is a six-year issue that pays fixed interest payments of C% of the *nominal* value on an annual basis. In the sixth year there is a final interest payment and the loan proceeds represented by the bond are also paid back, known as the maturity proceeds. The amount raised by the bond issuer is dependent on the price of the bond at issue, usually *par* or 100 per cent of face value, which we have labelled here as the issue proceeds.

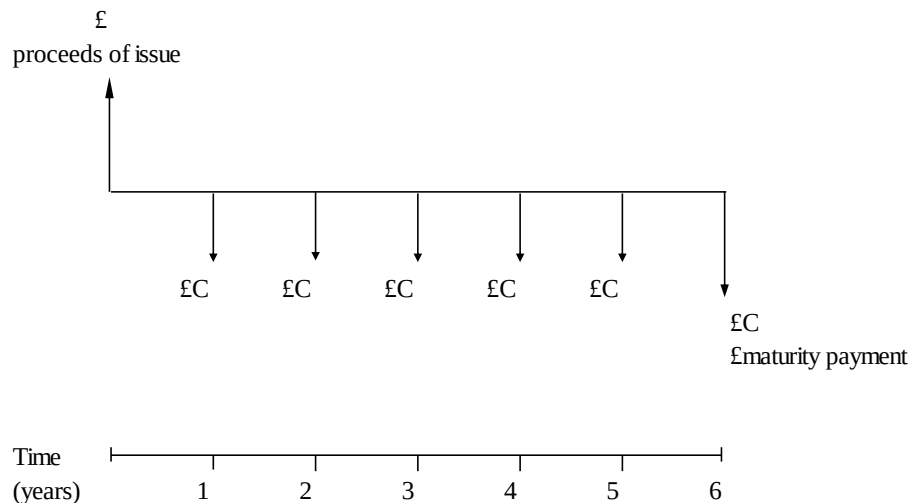


Figure 2.1 Cash flows associated with a six-year annual coupon bond

The upward facing arrow represents the cash flow paid and the downward facing arrows are the cash flows received by the bond investor. The cash flow diagram for a six-year bond that had a 5% fixed interest rate, known as a 5% *coupon*, would show interest payments of £5 per every £100 of bonds, with a final payment of £105 in the sixth year, representing the last coupon payment and the redemption payment. Again, the amount of funds raised per £100 of bonds depends on the price of the bond on the day it is first issued, and we will look further into this later. If our example bond paid its coupon on a semi-annual basis, the cash flows would be £2.50 every six months until the final redemption payment of £102.50.

Gilts and US government bonds, known as Treasuries, pay coupon every six month. Other bonds pay annual coupon or quarterly coupon.

Let us examine some of the key features of bonds.

Type of issuer. The primary distinguishing feature of a bond is its issuer. The nature of the issuer will affect the way the bond is viewed in the market. There are four issuers of bonds: sovereign governments and their agencies, local government authorities, supranational bodies such as the World Bank and corporations. Within the corporate bond market there is a wide range of issuers, each with differing abilities to satisfy their contractual obligations to investors. The largest bond markets are those of sovereign borrowers, the government bond markets.

Term to maturity. The *term to maturity* of a bond is the number of years after which the issuer will repay the obligation. The *maturity* of a bond refers to the date that the debt will cease to exist, at which time the issuer will redeem the bond by paying back to bondholders the *principal* or face value. The term to maturity is an important consideration in the make-up of a bond. It indicates the time period over which the bondholder can expect to receive the coupon payments and the number of years before the principal will be paid in full. The bond's *yield* is also depends on the term to maturity. Finally, the price of a bond will fluctuate over its life as yields in the market change and as it approaches maturity. As we will discover later, the *volatility* of a bond's price is dependent on its maturity; assuming other factors constant, the

longer a bond's maturity the greater the price volatility resulting from a change in market yields.

Principal and coupon rate. The *principal* of a bond is the amount that the issuer agrees to repay the bondholder on the maturity date. This amount is also referred to as the *redemption value*, *maturity value*, *par value* or *face amount*, or simply *par*. The *coupon rate* or *nominal rate* is the interest rate that the issuer agrees to pay each year. The annual amount of the interest payment made is called the *coupon*. The coupon rate multiplied by the principal of the bond provides the cash amount of the coupon. For example a bond with a 7% coupon rate and a principal of £1,000,000 will pay annual interest of £70,000. In the United Kingdom, United States and Japan the usual practice is for the issuer to pay the coupon in two semi-annual instalments. For bonds issued in European markets and the Eurobond market coupon payments are made annually. On rare occasions one will encounter bonds that pay interest on a quarterly basis. All bonds make periodic interest payments except for *zero-coupon bonds*. These bonds allow a holder to realise interest by being sold substantially below their principal value. The bonds are redeemed at par, with the interest amount then being the difference between the principal value and the price at which the bond was sold. We will explore zero-coupon bonds in greater detail later.

Currency. Bonds can be issued in virtually any currency. The largest volume of bonds in the global markets are denominated in US dollars; other major bond markets are denominated in euros, sterling and Japanese yen. There are also liquid markets in Australian, New Zealand and Canadian dollars, Swiss francs and other major currencies.

B. Bond Issuers

The core of any domestic capital market is the government bond market, which also forms the benchmark for all other borrowing. Figure 2.2 below illustrates UK gilt price and yield quotes as listed in the *Financial Times* for 28 August 2004.

£104.61 per £100 of stock today, we will receive £100 per £100 of stock on maturity. At first sight this appears to imply we will lose money, however we also receive coupon payments every six months, which for this bond is £2.50 per £100 nominal of stock. Also from figure 2.2 we see the change in price from the day before for each gilt, and in the case of the 5¾% 2009 the price was £0.07 higher than the previous day's closing price.

Government agencies also issue bonds. Such bonds are virtually as secure as government bonds. In the United States agencies include the Federal National Mortgage Association. Local authorities issue bonds as part of financing for roads, schools, hospitals and other capital projects.

Corporate borrowers issue bonds both to raise finance for major projects and also to cover ongoing and operational expenses. Corporate finance is a mixture of debt and equity and a specific capital project will often be financed as a mixture of both.

What bonds are NOT

Before proceeding with our discussion, let's consider a number of investments that are called "bonds" in the financial media and by financial advisors, but are *not* the bonds that we are talking about in this book.

Savings bonds

These are marketed by banks and building societies, but they are in fact savings accounts. The difference is that they have fixed interest rates or fixed terms, unlike instant access savings accounts, and so investors cannot access their money until the term is completed.

With-profits bonds

These are savings plans that are usually set for ten-year terms or longer. They are offered by life assurance companies. Savers invest a lump sum or regular monthly amounts into a plan that is invested in the "with-profits" fund of the life company, which are designed to smooth out the cyclical return patterns of the stock market. With-profit funds are overwhelmingly invested in equities.

Premium bonds

These are a glorified type of lottery, with the added advantage that the investor can get his investment back at any time. Premium bonds are entered in a prize draw, with a sliding scale of prizes.

Guaranteed income bonds

This is another product offered by life assurance companies, in which a lump sum is invested in a fixed term. The return is guaranteed during this term, and the original capital investment returned on maturity.

Baby bonds

These are offered by life companies and are another form of regular savings into a with-profits fund.

B. Types of bonds

The needs of about 95% of private investors will be met by conventional bonds. But as we noted in chapter one, bonds come in a variety of shapes and sizes. Let's consider these here.

Conventional bonds

These are the most common type and have a fixed coupon and fixed maturity date. They are also known as *bullet*, *straight* or *vanilla* bonds.

Government bonds

These are bonds issued by governments and include the safest investments in the market, such as UK gilts, US Treasuries, German Bunds and so on.

Floating Rate Notes

The bond market is often referred to as the *fixed income* market, or the *fixed interest* market in the UK. Floating rate notes (FRNs) do not have a fixed coupon at all but instead link their interest payments to an external reference, such as the three-month bank lending rate. Bank interest rates will fluctuate constantly during the life of the bond and so an FRN's cash flows are not known with certainty. Usually FRNs pay a fixed margin or *spread* over the specified reference rate; occasionally the spread is not fixed and such a bond is known as a *variable rate note*. Because FRNs pay coupons based on the three-month or six-month bank rate they are essentially money market instruments and are treated by bank dealing desks as such.

Index-linked bonds

An index-linked bond has its coupon and redemption payment, or possibly just either one of these, linked to a specified index. When governments issue Index-linked bonds the cash flows are linked to a price index such as consumer or commodity prices. Corporates have issued index-linked bonds that are connected to inflation or a stock market index.

Zero-coupon bonds

Certain bonds do not make any coupon payments at all and these are known as *zero-coupon bonds*. A zero-coupon bond or *strip* has only one cash flow, the redemption payment on maturity. If we assume that the maturity payment is say, £100 per cent, the issue price will be at a discount to par. Such bonds are also known therefore as *discounted* bonds. The difference between the price paid on issue and the redemption payment is the interest realised by the bondholder. As we will discover when we look at strips this has certain advantages for investors, the main one being that there are no coupon payments to be invested during the bond's life. Both governments and corporates issue zero-coupon bonds.

Corporate bonds

These are bonds issued by non-government entities and can be conventional bonds or a variation on a theme. Corporate bonds exhibit varying degrees of credit risk, because they are not risk-free like gilts. The yield payable on a corporate bond will depend on how risky its issuer is perceived in the market.

High yield bonds

These are also known as junk bonds and are issued by companies that have a *credit rating* below *investment grade*. This means that the company represents a higher risk for investors than usual, and so the yield on the bond is considerably higher than (say) government bonds or investment grade companies. They are, almost invariably, not suitable for private investors.

Eurobonds

These are also known as international securities, and are issued by governments and companies to investors around the world.² They are generally conventional bonds, but include a large class of bonds known as asset-backed securities. These are not generally suitable for private investors.

Bonds with embedded options

Some bonds include a provision that gives either the bondholder and/or the issuer an option to select early redemption of the bond. The most common type of option embedded in a bond is a *call feature*. A call provision grants the issuer the right to redeem all or part of the debt before the specified maturity date. An issuing company may wish to include such a feature as it allows it to replace an old bond issue with a lower coupon rate issue if interest rates in the market have declined. As a call feature allows the issuer to change the maturity date of a bond it is considered harmful to the bondholder's interests; therefore the market price of the bond at any time will reflect this. A bond issue may also include a provision that allows the investor to change the maturity of the bond. This is known as a *put feature* and gives the bondholder the right to sell the bond back to the issuer at par on specified dates. The advantage to the bondholder is that if interest rates rise after the issue date, thus depressing the bond's value, the investor can realise par value by *putting* the bond back to the issuer. A *convertible* bond is an issue giving the bondholder the right to exchange the bond for a specified amount of shares (equity) in the issuing company. This feature allows the investor to take advantage of favourable movements in the price of the issuer's shares.

Bond warrants

A bond may be issued with a *warrant* attached to it, which entitles the bond holder to buy more of the bond (or a different bond issued by the same borrower) under specified terms and conditions at a later date. An issuer may include a warrant in order to make the bond more attractive to investors. Warrants are often detached from their host bond and traded separately.

Asset-backed securities

These are bonds formed from pooling together a set of loans such as mortgages or car loans and issuing bonds against them. The interest payments on the original loans serve to back the interest payable on the asset-backed bond. They are not marketed to private investors and we will not be considering these instruments in this book.

² You will sometimes read that Eurobonds are bonds that are issued in a different currency to that of the issuer, but this is incorrect. The true definition is a bond that is issued in the international capital markets, usually by a syndicate of banks, and "cleared" (settled) in the either of the two main clearing systems for Eurobonds, Euroclear and Clearstream.

B. The queasy stuff: price and yield

In the preface we noted Professor Steven Hawking's belief that for each mathematical equation in a book, the readership is cut by half. Regrettably then, in this chapter we're going to lose some of our readers. But we won't forget Professor Hawking's dictum: we'll keep the numbers stuff to a minimum. In fact, we're going to prove once and for all that it is possible to understand bonds without having to get into the maths.

C. Financial Arithmetic

We have already described bonds as a package of cash flows. That is, a bond is a contract that states that the borrower (issuer) of the funds will pay a specified rate of interest on the borrowed money for the life of the deal, and then repay the loan in full on the maturity date. In other words, a bond is a package of *future cash flows*. What does one pay now for a package of future cash flows, if one was offered one?

Essentially, one would like to pay the fair price for such a package. But what is its fair value? Actually, we are not going too far from the truth if we say that the fair price of anything is the price at which a buyer and seller can agree. However both parties need to be fully informed. For bonds, it's all concerned with *discounting* and *present value*. We wish to know the present value of a package of future cashflows.

That's what we're going to talk about now...trust me, it's a piece of cake.

Discounting and Present Value

The principles of *compound interest* have long been used to illustrate that £1 received today is not the same as £1 received at a point in the future. Faced with a choice between receiving £1 today or £1 in one year's time, we would pick the £1 now – because there is no uncertainty that we will have it in our hands! However if we were offered the £1 in the future together with some interest, then we might accept to forgo the £1 now, just as long the interest rate offered was sufficient for us.

Say the rate of interest was 10% per annum. Our choice would be between £1 today and £1 plus 10p - the interest on £1 for one year at 10% per annum. But the further one goes into the future, the greater will be our requirement to be compensated for interest foregone, because of the effect of *compounding*. This is where we earn interest on our interest, provided it is added to the original investment and not spent, and it is an important concept for private investors

In compounding we seek to find a *future value* given a *present value*, a *time period* and an *interest rate*. If £100 is invested today (at time t_0) at 10%, then one year later (t_1) the investor will have $£100 \times (1 + 0.10) = £110$. If she leaves the capital and interest for another year she will have at the end of year 2 (t_2) :

$$\begin{aligned} & £110 \times (1 + 0.10) \\ = & £100 \times (1 + 0.10) \times (1 + 0.10) \\ = & £100 \times (1 + 0.10)^2 \\ = & £121. \end{aligned}$$

The outcome of the process of compounding is the *future value* of the initial amount. Therefore we can use the following expression

$$FV = PV(1+r)^n \quad (2.1)$$

where

FV is the future value
 PV is initial outlay or *present value*
 r is the periodic rate of interest (expressed as decimal)
 n is the number of periods for which the sum is invested

The expression (x.1) assumes *annual compounding*. Where semi-annual or quarterly compounding takes place, the equation needs to be modified slightly. We will receive a bit more interest the greater the frequency of compounding

Discounting

But what about bonds? They are a package of future cash flows. In other words, it is not the future value we want to find with them, because we already know what that is! If I want to buy a bond, I actually need to find its price today, which is its *present value* (PV) of a known future sum. Easy...we simply reverse the expression at (2.1), as shown:

$$PV = \frac{FV}{(1+r)^n} \quad (2.2)$$

where the letters PV , r and n mean the same things as before.

Example 2.1

Debbie needs to have £1000 in three years time to pay for a trip round the world. She can invest at 9%. How much does she need to invest now?

To solve this we require the present value of £1000 received in three years time, given that the interest rate (the discount rate) is 9%. This is given by:

$$\begin{aligned} PV &= \frac{1000}{(1+0.09)^3} \\ &= \frac{1000}{1.295} = £772.20 \end{aligned}$$

In other words, Debbie needs to invest about £773 now.

Compounding More than Once a Year

When interest is compounded more than once a year, the formula for calculating present values must be modified, as shown below.

$$PV = \frac{C_n}{\left(1 + \frac{r}{m}\right)^{mn}} \quad (2.3)$$

where C_n is the cash flow at the end of year n , m is the number of times a year interest is compounded, and r is the rate of interest as before. Therefore the present value of £100 to be received at the end of year 3, at a rate of interest rate of 10% compounded quarterly, is :

$$\begin{aligned} PV &= \frac{100}{\left(1 + \frac{.10}{4}\right)^{(4)(3)}} \\ &= £74.36 \end{aligned}$$

C. Pricing Bonds

So now we can easily determine what the price of a bond is...it is its *present value*. In other words, its price is equal to the present value of all its expected cash flows. A vanilla bond pays a fixed rate of interest (coupon) annually, semi-annually or quarterly. The *fair price* of such a bond is given by the discounted present value of all its cash flows, which we know. We obtain the present value by discounting the cash flows using a discount rate, which is in fact the bond's *gross redemption yield* or *yield-to-maturity* and is the rate r that appears in the expression (2.6).

$$\begin{aligned} P &= \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \dots + \frac{C}{(1+r)^{2T-1}} + \frac{C}{(1+r)^{2T}} + \frac{M}{(1+r)^{2T}} \\ &= \sum_{t=1}^{2T} \frac{C}{(1+r)^t} + \frac{M}{(1+r)^{2T}} \end{aligned} \quad (2.6)$$

where

- P is the fair price of the bond
- C is the annual fixed coupon payment
- M is the par value of the bond (usually 100)
- T is the number of *complete* years to maturity
- r is the market-determined discount rate or required rate of return for the bond.

For a bond that pays coupons semi-annually, like gilts, the expression is modified as shown below:

$$P = \frac{C/2}{(1 + \frac{1}{2}r)} + \frac{C/2}{(1 + \frac{1}{2}r)^2} + \dots + \frac{C/2}{(1 + \frac{1}{2}r)^{2T-1}} + \frac{C/2}{(1 + \frac{1}{2}r)^{2T}} + \frac{M}{(1 + \frac{1}{2}r)^{2T}}$$

$$= \sum_{t=1}^{2T} \frac{C/2}{(1 + \frac{1}{2}r)^t} + \frac{M}{(1 + \frac{1}{2}r)^{2T}}.$$

(2.7)

Looks scary doesn't it! But trust me, it ISN'T!

All (2.7) is saying is that we calculate the present value of each coupon, as well as the principal payment, and add all these amounts together. The funny squiggly sign, the Greek letter “sigma”, just means “add up”.

Of course, from above we see that the expression (x.6) contains two unknowns – the price and the redemption yield. We then remember from “O” level maths levels that we cannot solve two unknowns from only one equation. So we have to solve it iteratively. Can we be bothered with that? No, not really, it's much easier if we use a financial calculator such as a Hewlett Packard, or something like Excel. The appendix at the back shows how to do it using Excel. Now that really is easy!

Clean and Dirty Bond Prices: Accrued Interest

The bond price given by (2.6) is what is known as the *clean price*. When we buy a bond, we pay what is known as the *dirty price*. After a buy or sell transaction, there is a day when the buyer hands over her cash and the seller hands over the bond. This will NOT be the same day that the transaction or *bargain* takes place: it will be one, two, three or more days later and is known as the *settlement date*.

Unless the settlement date is the same day as the coupon date, the clean price will always be different to the dirty price. This is because we need to allow for what is known as accrued interest.

The clean price excludes coupon interest that has accrued on the bond since the last coupon payment. All bonds accrue interest on a daily basis, so that even if a bond is held for only one day, interest will have been earned by the bondholder. The price that is actually paid for the bond in the market is the *dirty price* (or *gross price*), which is the clean price plus accrued interest. In other words the accrued interest must be added to the quoted price to get the total consideration for the bond. Accrued interest compensates the seller of the bond for giving up all of the next coupon payment even though she will have held the bond for part of the period since the last coupon payment. The clean price for a bond will move with changes in market interest rates; assuming that this is constant in a coupon period, the clean price will be constant for this period. However the dirty price for the same bond will increase steadily from one interest payment date until the next one. On the coupon date the clean and dirty prices are the same and the accrued interest is zero. Between the coupon payment date and the next *ex dividend* date the bond is traded *cum dividend*, so that the buyer gets the next coupon payment. The seller is compensated for not receiving the next coupon payment by receiving accrued interest instead. This is positive and increases up to the next ex dividend date, at which point the dirty price falls by the present value of the amount of the coupon payment. The dirty price at

this point below the clean price, reflecting the fact that accrued interest is now negative. This is because after the ex dividend date the bond is traded “ex dividend”; the seller not the buyer receives the next coupon and the buyer has to be compensated for not receiving the next coupon by means of a lower price for holding the bond.

Not all markets have an ex-dividend period, and trade cum dividend right up to the coupon date. Gilts do have an ex-div period.

In an ideal word, to calculate accrued interest we would take the number of days we’d held the bond (actual days) and divided it by 365. We would then multiply the bond coupon by this fraction. In other words, the formula *should* be

$$AI = C \times \frac{act}{365}. \quad (2.8)$$

In fact it isn’t, not for a lot of bonds anyway! If you come across this formula in a book, be aware that it will give you a good approximation but *not* the precise value. Gilts, Treasuries and a lot of euro government bonds calculate accrued interest on what is known as an *act/act* basis, but unless we are dealing in large sums, it is not really an issue for us.

Example 2.2

Accrued interest calculation for 7% Treasury 2002

This bond has coupon dates of 7 June and 7 December each year. Say that £100 nominal of the bond is traded for value 27 August 1998, at a clean price of 98.25. What is accrued interest on the value date and the total consideration?

On the value date 81 days has passed since the last coupon date. Under the old system for gilts, *act/365*, the calculation was:

$$7 \times \frac{81}{365} = 1.55342.$$

Under the current system of *act/act*, which came into effect for gilts in November 1998, the accrued calculation uses the actual number of days between the two coupon dates, giving us:

$$7 \times \frac{81}{183} \times 0.5 = 1.54918.$$

We add this to the clean price to obtain a dirty price of £98.799 (rounded to 98.80) per £100 face value of bond.

Simple!

C. Bond yields

There are various types of yield measure used in the market. We need only concern ourselves with two, the simple yield or *running yield* and the *redemption yield*.

Running yield

The simplest measure of the yield on a bond is the *current yield* (or *flat yield*, *interest yield* or *running yield*). This is defined as

$$rc = \frac{C}{P} \times 100$$

(x.2)

where

rc is the current yield
 C is the coupon rate
 P is the clean price.

The current yield is useful as a “rough-and-ready” interest rate calculation; it is often used to estimate the cost of or profit from a short-term holding of a bond. For example if we buy a bond, hold it for several months and then sell it, the running yield is what we would have earned during this period. It is a good approximation, and enables us to compare the return on the bond holding with a bank account or other return.

Figure 2.2 showed the running yield for each gilt as well as the redemption yield.

Example 2.3 **Running yield**

If the clean price of a bond is 98.76 and the coupon is 6.00% then the current yield is

$$rc = \frac{6.00}{98.76} \times 100$$

$$= 6.075\%.$$

Yield to Maturity

The *yield to maturity* (YTM) or *gross redemption yield* is the most frequently used measure of return from holding a bond. It takes into account the pattern of coupon payments, the bond’s term to maturity and the capital gain (or loss) arising over the remaining life of the bond. We know the expression for it already – it is the rate r given in (x.6) earlier. This uses r to discount the bond’s cash flows back to the next coupon payment and then discounts the value at that date back to date t . In other words r is the *internal rate of return* (IRR) that equates the value of the discounted cash flows on the bond to the current dirty price of the bond (if date t is the current date). The internal rate of return is the discount rate which, if applied to all of the cash flows will solves for a number that is equal to the dirty price of the bond (its

present value). By assuming that this rate will be unchanged for the reinvestment of all the coupon cash flows, and that the instrument will be held to maturity, the IRR can then be seen as the yield to maturity.

Note that the redemption yield as calculated by equation (x.6) is the *gross redemption yield*, the yield that results from payment of coupons without deduction of any withholding tax. The *net redemption yield* is obtained by multiplying the coupon rate by (1 - marginal tax rate). The net redemption yield is lower than the gross redemption yield.

What does the redemption yield tell us?

While redemption yield is the most commonly used measure of yield, it has one major disadvantage. The disadvantage is that implicit in its calculation is the assumption that each coupon payment as it becomes due is re-invested at the rate r . This is clearly unlikely, due to the fluctuations in interest rates over time and as the bond approaches maturity. In practice the measure itself will not equal the actual return from holding the bond, even if it is held to maturity. That said, the market standard is to quote bond returns as yields to maturity, bearing the key assumptions behind the calculation in mind, so that we can compare yields across different bonds and investments.

Yield on a zero-coupon bond

Zero-coupon bonds, sometimes known as *strips*, have only one cash flow, the redemption payment on maturity. Typically this payment will be par (100) and hence the bond is sold at a discount to par and trades at a discount during its life. For a bond with only one cash flow it is obviously not necessary to use (x.6) and instead we can use (x.x) as stated below.

$$P = \frac{C}{(1+r)^n} \quad (\text{x.x})$$

Equation (x.x) above substitutes n for S ; n is the number of years to the bond's redemption. We can re-arrange the equation for the yield, r .

$$r = \left(\frac{C}{P} \right)^{\frac{1}{n}} - 1 \quad (2.13)$$

Index-linked yields

In certain markets including the US and UK, the government and certain companies issue bonds that have either or both of their coupon and principal linked to a price index, such as the retail price index (RPI), the main measure of inflation. If we wish to calculate the yield on such *TIPS* or *index-linked bonds* it is necessary to make forecasts of the relevant index, which are then used in the yield calculation. As an example we can use the case of index-linked government bonds which were first introduced in the UK in March 1981. Both the principal and coupons on UK index-linked gilts are linked to the RPI and are therefore designed to give a constant *real*

yield. Most of the index-linked stocks that have been issued by the UK government have coupons of 2 or 2½ per cent. Apparently this was because of the fact that the long-run real rate of return on the UK capital stock is about 2½ per cent.

Both the coupon and the principal on index-linked gilts are scaled up by the ratio of two values of the RPI. The denominator of this ratio is known as the *base RPI* and is the value of the RPI eight months prior to the month of the issue of the bond. The numerator is the value of the RPI eight months prior to the month of the relevant coupon or principal payment.

The semi-annual coupon payment is given by 2.19 below.

$$\text{Coupon payment} = (C / 2) \times \frac{RPI_{m-8}}{RPI_b} \quad (2.19)$$

The principal repayment is given by 2.20.

$$\text{Principal repayment} = 100 \times \frac{RPI_{M-8}}{RPI_b} \quad (2.20)$$

where C is the *real* annual coupon payment and where

RPI_b is the value of RPI eight months prior to the month of issue (the base RPI)
 RPI_{m-8} is the value of RPI eight months prior to the month in which coupon is paid
 RPI_{M-8} is the value of RPI eight months prior to the month of bond redemption

The eight-month lag in the RPI is used for the following reason. The authorities require to know each coupon payment six months before it is paid in order to determine the interest accruing between coupon payments. The two additional months are explained by the one-month delay in publishing the RPI (for example, June's RPI is not published until July) and an allowance of one month to make the relevant calculations. It is also important to take into account any rebasing of the RPI. The index was last rebased in January 1987, to 100 from the January 1974 value of 394.5.

Example 2.4

Index-linked Treasury 2% 1988 was issued in March 1982 and matured on 30 March 1988. The base RPI was 297.1 (that is, the RPI for July 1981); the RPI for July 1987 was 101.8. Thus we can calculate the money value of the final coupon payment and principal repayment on 30 March 1988 as shown below.

$$\text{Coupon payment} = (2 / 2) \times \frac{394.5}{297.1} \times \frac{101.8}{100.0} = \text{£}1.3517$$

$$\text{Principal repayment} = 100 \times \frac{394.5}{297.1} \times \frac{101.8}{100.0} = \text{£}135.1736$$

The interest accruing during the last coupon period (30 September 1987 - 30 March 1988) can be calculated once we know the final coupon payment, given below.

$$1.3517 \times \frac{\text{No. of days accrued}}{(365 / 2)}$$

There are two kinds of index-linked yields to maturity : the money (or nominal) yield, and the real yield. The money yield requires forecasts of all future cash flows from the bond. This in turn requires forecasts of all the relevant future RPIs. The commonest way of doing this is to take the latest available RPI and assume a constant inflation rate thereafter.

The forecast for the first relevant future RPI is determined using 2.21;

$$RPI_1 = RPI_0 \times (1 + \tau)^{m/12} \quad (2.21)$$

where

RPI_0 is the latest available RPI

τ is the forecast of the annual inflation rate

m is the number of months between RPI_0 and RPI_1

Suppose for instance that the bond pays coupons every March and September. The relevant months for forecasting the RPI are eight months prior to March and September, namely July and January. If the latest available RPI is for October, then we are attempting to make a forecast for the RPI the following January, in which case $m = 3$. The forecast for each subsequent relevant RPI is found using 2.22

$$RPI_{j+1} = RPI_1 \times (1 + \tau)^{j/2} \quad (2.22)$$

where j is the number of semi-annual forecasts after RPI_1 (which was the forecast for the RPI in January). For example if the October RPI is 102.8 and an annual inflation rate of 4.5 per cent is expected, then the forecast for the RPI for the following January is

$$\begin{aligned} RPI_1 &= 102.8 \times (1.045)^{3/12} \\ &= 103.9 \end{aligned}$$

and for the January following that it is

$$\begin{aligned} RPI_3 &= 103.9 \times (1.045) \\ &= 108.6 \end{aligned}$$

The *money yield* (rm) is calculated by solving the following equation, assuming that the calculation is made on a coupon payment date, so that accrued interest is zero :

$$P_d = \frac{(C/2)(RPI_1 / RPI_b)}{(1 + \frac{1}{2} rm)} + \frac{(C/2)(RPI_2 / RPI_b)}{(1 + \frac{1}{2} rm)^2} + \dots + \frac{(\frac{C}{2} + M)(RPI_S / RPI_b)}{(1 + \frac{1}{2} rm)^S}$$

(2.23)

where S is the number of coupon payments before redemption.

The *real yield* (ry) is related to the money yield through equation 2.24, which was first described by Fisher in *Theory of Interest* (1930).

$$(1 + \frac{1}{2} ry) = (1 + \frac{1}{2} rm) / (1 + \tau)^{\frac{1}{2}} \quad (2.24)$$

For example if the money yield is 7 per cent and the forecast inflation rate is 4.5 per cent, then the real yield is found from 2.24 as shown.

$$ry = \left\{ \frac{[1 + \frac{1}{2}(0.07)]}{[1 + (0.045)]^{\frac{1}{2}}} - 1 \right\} \times 2$$

$$\cong 0.025 \quad (2.5 \text{ per cent})$$

Equation 2.23 is used to show that the money yield is the appropriate discount rate for discounting money or nominal cash flows, and can be rearranged to show that the real yield is the appropriate discount rate for discounting real cash flows.

To compare returns between index-linked bonds and conventional bonds it is necessary to calculate the *break-even inflation rate*. This is the inflation rate that makes the money yield on an index-linked bond equal to the redemption yield on a conventional bond of the same maturity. Suppose that the redemption yield on a conventional bond is 5 per cent and that the real return on an index-linked bond is 2.5 per cent. Using 2.24 this gives us a break-even inflation rate of :

$$\tau = \left\{ \frac{[1 + \frac{1}{2}(0.05)]}{[1 + \frac{1}{2}(0.025)]} \right\}^2 - 1$$

= 0.02484 (2.48 %).

If the expected rate of inflation is higher than the break-even rate of inflation, investors will prefer the index-linked bond, and vice-versa.

The price / yield relationship

The foregoing analysis have illustrated a fundamental property of bonds, namely that an upward change in the price results in a downward move in the yield, and vice-versa. This is of course immediately apparent since the price is the present value of the cash flows; as the required yield for a bond say, decreases the present value and hence the price of the cash flow for the bond will increase. It is also reflects the fact that for plain vanilla bonds the coupon is fixed, therefore it is the price of the bond that will need to fluctuate to reflect changes in market yields. It is useful sometimes to plot the relationship between yield and price for a bond. A typical price / yield profile is represented graphically at figure 2.3, which shows a *convex* curve.

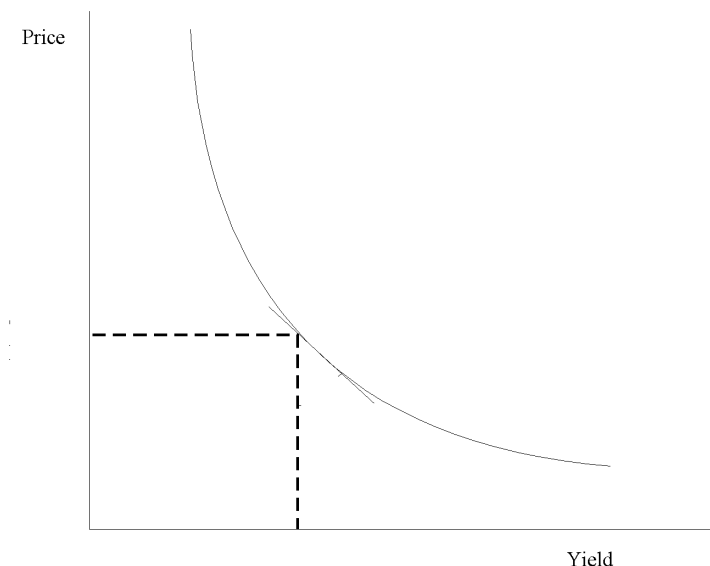


Fig 2.3 The bond price / yield relationship

As we have said, for a plain vanilla bond with a fixed coupon, the price is the only variable that can change to reflect changes in the market environment. When the coupon rate of a bond is equal to the market rate, the bond price will be par (100). If the required interest rate in the market moves above a bond's coupon rate at any point in time, the price of the bond will adjust downward in order for the bondholder to realise the additional return required. Similarly if the required yield moves below the coupon rate, the price will move up to equate the yield on the bond to the market rate. As a bond will redeem at par, the capital appreciation realised on maturity acts as compensation when the coupon rate is lower than the market yield.

The price of a bond will move for various reasons, including the market-related ones noted here :

- when there is a change in the yield required by the market, either because of changes in the base rate or a perceived change in credit quality of the bond issuer (credit considerations do not affect developed country government bonds);
- when there is a change because as the bond is approaching maturity, its price moves gradually towards par;
- when there is a change in the market-required yield due to a change in the yield on comparable bonds.

Bond prices also move for liquidity reasons and normal supply-and-demand reasons, for example if there is a large amount of a particular bond in issue it is easier to trade the bond; also if there is demand due to a large customer base for the bond. Liquidity is a general term used here to mean the ease with which a market participant can trade in or out of a position. If there is always a ready buyer or seller for a particular bond, it will be easier to trade in the market.

So, we know now the following:

- If the yield rises, the price of the bond goes down
- If the yield falls, the price of the bond goes up
- If the price of the bond is below par (100), the coupon is lower than the yield, and if the price is above par, then the coupon is above the yield.

Good stuff, lets move on...

B. More queasy stuff: a (thankfully!) brief word on duration

It's worth spending just a bit of time on the subject of duration, and the concept of interest rate risk. Private investors don't need to get too bogged down in this area, just as long as they are aware of it.

One of the key identifying features of a bond is its term to maturity. However this does not tell us the full story on the timing of the bond's cash flows, or its price behaviour in the market compared to other bonds. All bonds, except zero-coupon bonds, pay part of their total return during their lifetime, in the form of coupon interest, so that the term to maturity does not reflect the true period over which the bond's return is earned. Also if we wish to gain an idea of the trading characteristics of a bond, and compare this to other bonds of say, similar maturity, the term to maturity is insufficient and so we need a more accurate measure.³

³ By "trading characteristics" we mean how it's price will change as a result of a change of market interest rates, compared to how the price of other bonds will change for the same movement in interest rates.

C. Revisiting the bond price / yield relationship

Earlier we showed how for a plain vanilla bond, because the coupon is fixed, the price is the only parameter of the bond's make-up that can change in response to a change in market interest rates. Hence a bond's price will move in the opposite direction to a move in market interest rates, where the latter has triggered a change in the required yield for that bond. The extent of the change in bond price will depend on how sensitive the bond is to interest rate changes. This is different for different bonds. For example, figure 2.4 shows the price/yield relationship for three bonds of differing maturities and coupons: this illustrates how the nature of the convex relationship between price and yield is different for each bond.

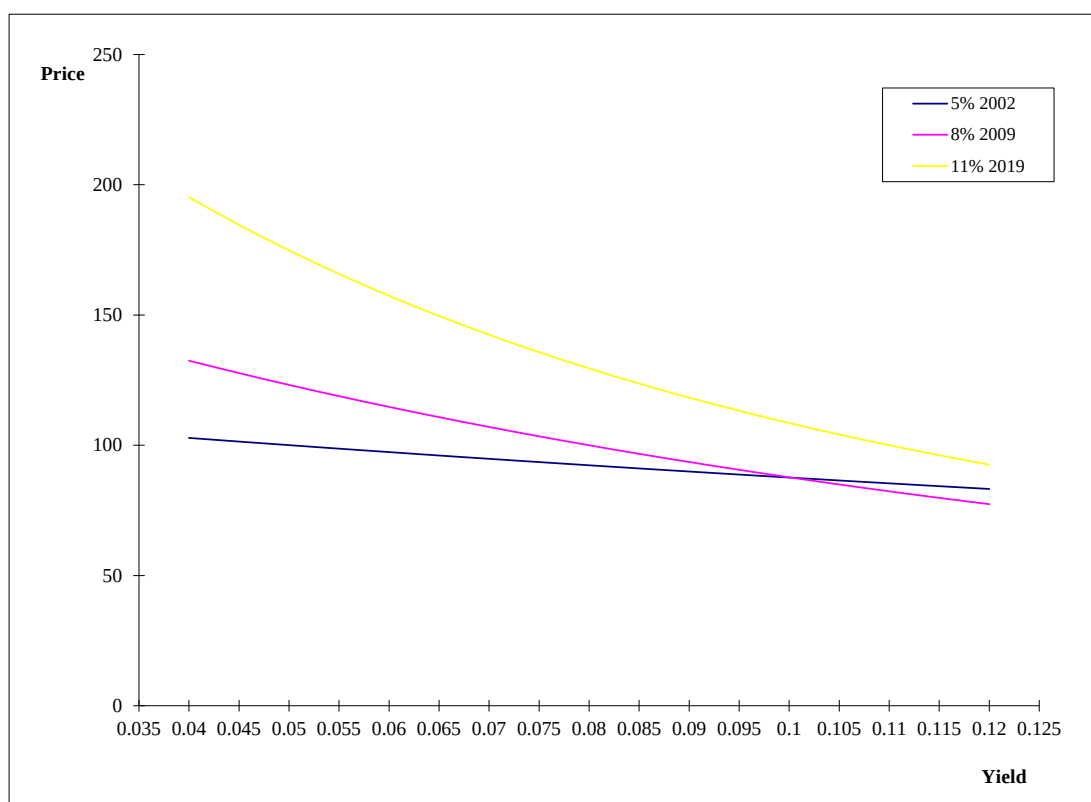


Figure 2.4 Price / yield relationship for three hypothetical bonds

From figure 2.4 note that the change in price for a given change in interest rates (see the change along the x-axis) is not uniform for an upward or downward change in yield.

If we measure the change along the price axis for a fixed change in the yield axis, we can see most obviously for the 11% 2019 bond that the price increase is greater than the price decrease for a given change in yield. This is observed for all bonds where the price / yield relationship is convex. We can also see from figure 7.1 that the 5% 2002 bond is less convex than the 11% 2019 bond; so, while the property of upward price changes outstripping downward price changes still holds, it is less obvious than for the longer-dated bond.

So what is the significance of this? Essentially, that as private investors we should be aware that certain bonds are more “risky”, from an interest-rate change point of view, than others. If we buy a bond with the intention of holding it until maturity, then we

will not be concerned with its level of interest rate risk. But we may be more comfortable holding bonds of lower interest-rate risk. We get an idea of the extent of this risk from duration or modified duration.

C. Duration

A plain vanilla coupon bond pays out a proportion of its return during the course of its life, in the form of coupon interest. If we were to analyse the properties of a bond, we should conclude quite quickly that its maturity gives us little indication of how much of its return is paid out during its life, nor any idea of the timing or size of its cash flows, and hence its sensitivity to moves in market interest rates. For example, if comparing two bonds with the same maturity date but different coupons, the higher coupon bond provides a larger proportion of its return in the form of coupon income than does the lower coupon bond. The higher coupon bond provides its return at a faster rate; its value is theoretically therefore less subject to subsequent fluctuations in interest rates.

We may wish to calculate an average of the time to receipt of a bond's cash flows, and use this measure as a more realistic indication of maturity. However cash flows during the life of a bond are not all equal in value, so a more accurate measure would be to take the average time to receipt of a bond's cash flows, but weighted in the form of the cash flows' present value. This is, in effect, *duration*. We can measure the speed of payment of a bond, and hence its price risk relative to other bonds of the same maturity by measuring the average maturity of the bond's cash flow stream. Bond analysts use duration to measure this property (it is sometimes known as *Macaulay's duration*, after its inventor, who first introduced it in 1938).⁴ Duration is the weighted average time until the receipt of cash flows from a bond, where the weights are the present values of the cash flows, measured in years. At the time that he introduced the concept, Macaulay used the duration measure as an alternative for the length of time that a bond investment had remaining to maturity.

We can illustrate Macaulay duration with a simple example, using an hypothetical 5-year bond with precisely five years to maturity and a coupon of 8 per cent. Assume that the bond is priced at par, giving a yield to maturity of 8 per cent. The bond's cash flows are shown in table 2.1, along with a diagram of the timing of cash flows at figure 2.5.

⁴ Macaulay, F., *Some theoretical problems suggested by the movements of interest rates, bond yields and stock prices in the United States since 1865*, National Bureau of Economic Research, NY 1938. This is actually quite a fascinating read and can be bought today, published by the RISK Classics Library.

8% five-year bond, priced at par			
Cash flow	Present Value*	Timing (t)	PV x t
8.00	7.41	1	7.41
8.00	6.86	2	13.72
8.00	6.35	3	19.05
8.00	5.88	4	23.52
108.00	73.50	5	367.51
	100.00		431.21
* Calculated as $C/(1+r)^t$			
Duration is the sum of all PV*t divided by P, sp 431.21/100 or 4.31 years			

Table 2.1 Example of duration calculation

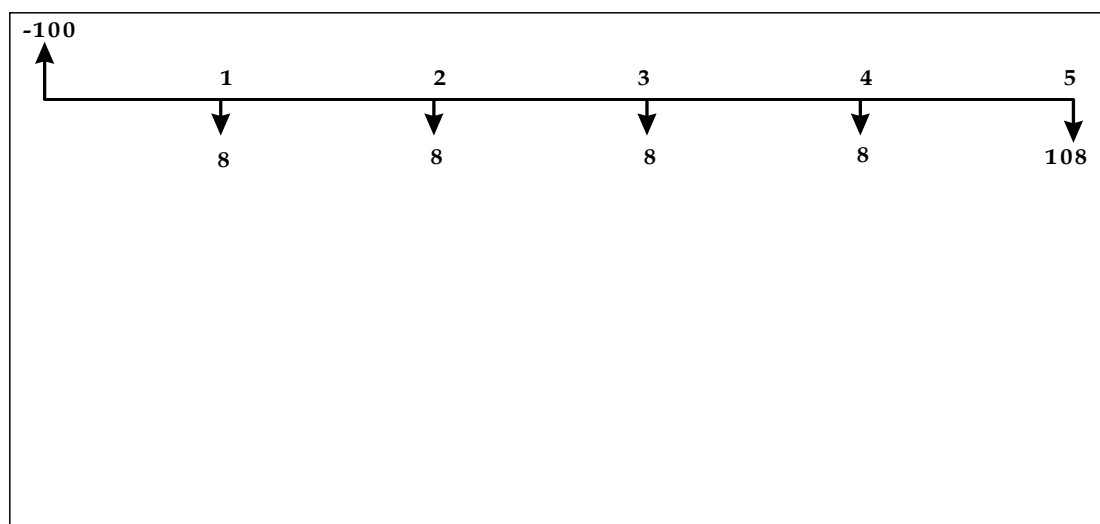


Figure 2.5 Receipt of cash flows for 8% 5-year bond

The present value of each cash flow is calculated in the normal way, hence for the period 2 cash flow of £8 the present value is $8 / (1.08)^2$, which gives us 6.859.

So we see that duration is then calculated using 2.25 below.

$$D = \frac{\sum (\text{present value of cash flow} \times \text{time to cash flow})}{\sum (\text{present value of cash flow})} \quad (2.25)$$

Mathematically this is written as (2.26) below.⁵

$$D = \frac{\sum_{n=1}^N \frac{nC_n}{(1+r)^n}}{P} \quad (2.26)$$

⁵ This assumes the bond has a whole number of interest periods to maturity; if this is not the case we round down to the nearest whole number

where

D is the Macaulay duration

C is the bond cash flow at time n

P is the present value of the n -period cash flow, discounted at the current yield to maturity

n is the number of interest periods.

Note that the denominator, as the sum of all the present values of the cash flows, is in fact the price of the bond.

For a semi-annual coupon bond, the cash flows are discounted at half the current yield to maturity. Equation 7.2 solves for Macaulay duration in terms of the number of interest periods; we divided this by the number of interest periods per year (either 1 or 2 for annual or semi-annual coupon bonds) to obtain the Macaulay duration in years.

In our illustration of the 8% five-year bond, shown at table 2.dur above, duration is calculated as $431.21 / 100$, which is equal to 4.31 years. This implies that the average time taken to receive the cash flows on this bond is 4.31 years. This is shown in figure 2.6 in our “duration fulcrum”, with 4.31 years being the time to the pivot, marked from A to B. The coupons are shown as “C”, and these diminish progressively as their present value decreases.

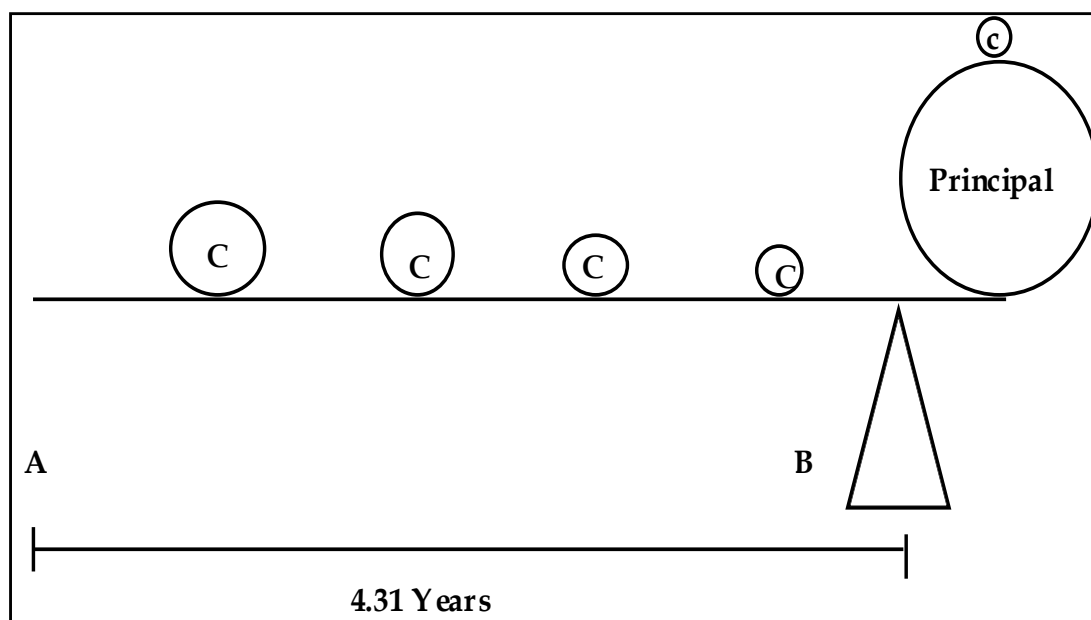


Figure 2.6 The Duration Fulcrum

For a zero-coupon bond, where the present value of the coupon payments is zero because there are no coupon payments, the Macaulay duration can be shown to be equal to the number of interest periods remaining in the bond’s life. If the bond is

trading in a bond market with an annual coupon convention, the Macaulay duration is therefore equal to the number of years remaining to maturity.

In the appendix we show how you can use Microsoft Excel to calculate a bond's duration. It is a very straightforward procedure...

Original Macauley duration has very little practical value to investors or traders alike, however from it can be derived a useful measure of bond price sensitivity to interest rates⁶ known as *modified duration*. This is given as

$$MD = \frac{D}{(1+r)} . \quad (2.27)$$

Modified duration is clearly related to duration, but is a much more useful measure of risk. We can use it to indicate that, for small changes in yield, a given change in yield results in an inverse change in bond price. The modified duration value of a bond shows how much the price of the bond changes for a 1% change in its yield. At this stage, I must point out that the duration and modified duration concept assume a number of factors, all of which are useless for the real world, most notably the assumption of a *parallel shift* in yields, but we don't have to worry about that here...

C. Properties of duration

Let us examine some of the properties of duration. A bond's duration is always less than its maturity. This is because some weight is given to the cash flows in the early years of the bond's life, which brings forward the average time at which cash flows are received. In the case of a zero-coupon bond, there is no present value weighting of the cash flows, for the simple reason that there are no cash flows, and so duration for a zero-coupon bond is equal to its term to maturity. Duration varies with coupon, yield and maturity. The following three factors imply *higher* duration for a bond :

- the lower the coupon;
- the lower the yield;
- broadly, the longer the maturity.

Duration increases as coupon and yield decrease. As the coupon falls, more of the relative weight of the cash flows is transferred to the maturity date and this causes duration to rise. Because the coupon on index-linked bonds is generally much lower than on vanilla bonds, this means that the duration of index-linked bonds will be much higher than for vanilla bonds of the same maturity. As yield increases, the present values of all future cash flows fall, but the present values of the more distant cash flows fall relatively more than those of the nearer cash flows. This has the effect of increasing the relative weight given to nearer cash flows and hence of reducing duration.

⁶ Referred to as *interest rate sensitivity* or *interest rate risk*.

So what is the upshot of all this? It is this: if we are intending to hold a bond for less than its full term to maturity (and given that investment decisions must always be flexible, to react to new information or a change in our personal circumstances), if we think that base interest rates are going to rise, we should hold bonds of lower duration, *all else being equal*. In other words, it is okay to compare duration of different gilts, because they have the same (zero) credit risk. We should never, as private investors, hold a bond for duration reasons only. Put it this way: if I think interest rates are going to fall, and I have already decided to invest some of my funds in gilts, then there is no harm in buying a longer-dated gilt. If I think rates are going to rise, I should get a shorter-dated gilt. But either way, I was going to buy a gilt: I'm just letting the duration concept influence my choice of *which* gilt to buy.

APPENDIX

Using Excel to calculate bond yield and duration

Excel worksheet showing gross redemption yield calculation and duration calculation for 5% 2012 semi-annual coupon conventional bond, settlement date 5 November 2002. The duration calculation uses the bond yield in the formula, which is cell D12, and which was calculated first.

	Yield to maturity calculation	
	Column C	Description
Row 3	05/11/2002	Settlement date
4	07/11/2012	Maturity date
5	5.00%	Coupon
6	98.95	Price
7	£100	Redemption value
8	2	Coupon frequency (see below)
9	1	Accrued interest basis (see below)
	Formula	Result
	=YIELD(C3, C4, C5, C6, C7, C8, C9)	0.0514
		To view the result as a percentage, adjust the format to "percentage" using the Format and Cells menus.
	Coupon frequency	
	Annual = 1, Semi-annual = 2, Quarterly = 4	
	Accrued interest	
	US 30/360 = 0	
	Actual / actual = 1	
	Actual / 360 = 2	
	Actual / 365 = 3	
	European 30/360 = 4	
	Duration calculation	
	Formula	
	=DURATION(C3, C4, C5, D12, C8, C9)	7.785688677
		The duration of the same bond above, with yield of 5.14%.